

EXP39 Vehicle Detection in a Video frame using OpenCV .

AIM:- Vehicle Detection in a Video frame

PROGRAM:

```
import cv2

cap = cv2.VideoCapture("traffic.mp4")

fgbg = cv2.createBackgroundSubtractorMOG2()

kernel = cv2.getStructuringElement(
    cv2.MORPH_ELLIPSE,
    (5, 5)
)

fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fourcc = cv2.VideoWriter_fourcc(*"mp4v")
output = cv2.VideoWriter(
    "Vehicle_Detection_Output.mp4",
    fourcc,
    fps,
    (width, height)
)

while True:
    ret, frame = cap.read()
    if not ret:
        break
    fgmask = fgbg.apply(frame)
    fgmask = cv2.morphologyEx(
        fgmask,
        cv2.MORPH_OPEN,
        kernel
    )
    fgmask = cv2.morphologyEx(
        fgmask,
```

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        cv2.MORPH_CLOSE,
        kernel
    )
contours, _ = cv2.findContours(
    fgmask,
    cv2.RETR_EXTERNAL,
    cv2.CHAIN_APPROX_SIMPLE
)
for contour in contours:
    area = cv2.contourArea(contour)
    if area > 1000:
        x, y, w, h = cv2.boundingRect(contour)
        cv2.rectangle(
            frame,
            (x, y),
            (x + w, y + h),
            (0, 255, 0),
            2
        )
        cv2.putText(
            frame,
            "Vehicle",
            (x, y - 10),
            cv2.FONT_HERSHEY_SIMPLEX,
            0.6,
            (0, 255, 0),
            2
        )
output.write(frame)
cv2.imshow("Vehicle Detection", frame)
if cv2.waitKey(30) & 0xFF == 27:
    break
cap.release()
output.release()

```

```
cv2.destroyAllWindows()
```

```
print("Output saved as Vehicle_Detection_Output.mp4")
```

INPUT:

[traffic.mp4](#)

OUTPUT:

[Vehicle_Detection_Output.mp4](#)

RESULT:- Thus the program is executed successfully.